

Smartphones are crucial in modern society, and their effectiveness depends on how long their battery lasts. This often seems unpredictable, but it can be attributed to factors such as usage behavior, environmental conditions, and battery aging. Understanding how different factors affect battery state of charge (*SOC*) can help users optimize daily usage to extend battery life. The objective of our research is to develop a mathematical model that illustrates the behavior of the state of charge. The model constructed is a continuous-time function depending on user behavior, screen brightness, processor load, network activity, etc. The model produces realistic *SOC* values over a single day as well as over multiple cycles. The simulations confirm that the model follows the expected trends of battery degradation over time during daily usage patterns. Local sensitivity analysis quantifies how *SOC* is relatively impacted by fluctuations in each parameter. Under baseline conditions, brightness and processor load were found to be the most impactful. Our model's primary strength is its logical structure and interpretability. The framework is not smartphone-specific and can be adjusted to represent other battery-powered devices. Practical recommendations include reducing screen brightness, limiting high-load applications, and avoiding prolonged exposure to high temperatures.

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1 Introduction

The battery-powered smartphone is essential for daily life in modern society. Since its introduction in 1992 [1], the smartphone has evolved to support calling, messaging, gaming, photography, and soon, on-device AI. As of 2025, 98% of Americans owned a smartphone [2].

Smartphone users are familiar with the small battery icon at the top of their screen, which is usually accompanied by a percentage. This percentage is a measure called the State of Charge (SOC) and informs the user how much battery is left.

Most of today's smartphones are fueled by Lithium-ion (Li-ion) batteries. Regarded as the industry standard, Li-ion batteries offer high energy and power density while remaining lightweight, making them suitable for smartphones. The primary limitation of Li-ion battery use is degradation over time [3]. Two of the main contributors to battery degradation are age and temperature. There are chemical and physical properties within a battery that wear down with repetitive charging and discharging, and are further damaged by high temperatures[3].

As smartphone batteries degrade, the usability of the device and its value to the user decrease. Once the battery is below a certain capacity, the user is likely to replace the phone, which can be expensive. Efficient smartphone usage optimizes the time-to-empty for each cycle while minimizing long-term damage to the battery.

The goal of this research is to model the battery loss of a smartphone under realistic conditions. We also aim to understand the sensitivity of contributing factors to better learn how to minimize

battery depletion and, ultimately, its degradation.

2 Methods

2.1 Assumptions

To capture the dynamics of Li-ion battery drain, a continuous-time model that represents both active and inactive drain is proposed. The assumptions of this model are as follows:

1. Before the cycle count starts, the battery is charged to a value varying between 70% and 100%.
2. The phone is only charged at the end of every cycle.
3. Li-ion battery drain only depends on active power consumption and idle power consumption.
4. Li-ion battery degradation depends on temperature and implicitly on cycle count. Effects of aging due to calendar time are ignored
5. An Apple iPhone 16 on cellular data will be the smartphone used for the model, with realistic arbitrary values used for various parameters.
6. The user's routine is consistent during each cycle.

2.2 Creating the Model

At the basic conceptual level, the State of Charge, *SOC*, is measured by the percentage of the initial

capacity of the battery that has not been used. This relationship can be modeled by:

$$SOC(t, T) = SOC_0 - \frac{Q_u(t)}{Q_{ra}(T)} \quad (1)$$

Table 1: Eq. 1 Variables/Parameters

Symbol	Description	Units
SOC	State of charge	Percentage
Q_u	Charge used	Wh
Q_{ra}	Remaining battery capacity	Wh
t	Time	Hours
T	Temperature	Celsius
a	Cycle count	N/a

Where $Q_u(t)$ is a continuous-time function representing the amount of battery used and $Q_{ra}(T)$ is a function representing the remaining battery capacity. The remaining battery capacity is the maximum amount of energy that the battery can hold and is used to scale the charge used when obtaining the state of charge to be more representative of the battery's age.

$$Q_{ra}(T) = Q_{ra-1} \cdot f_{temp}(T) \quad (2)$$

Table 2: Eq. 2 Variables/Parameters

Symbol	Description	Units
Q_{r0}	Initial Battery Capacity	Wh
f_{temp}	Degradation Factor	N/a

This remaining capacity is modeled as the initial battery capacity (brand new) scaled by factors determined by the current cycle, a , and the external temperature, T , the phone is exposed to. These factors will be explored later.

To obtain Q_u , its derivative was formulated first. The rate of battery usage, $\frac{dQ_u}{dt}$, can be modeled as the sum of power consumption and passive battery drain.

$$\frac{dQ_u}{dt} = P_c + P_{idle} \quad (3)$$

Here, power consumption, P_c , is modeled as a combination of passive and active factors that increase battery load, and P_{idle} is a device-specific rate.

Although older batteries may appear to lose charge faster during idle periods, this behavior may be attributed to reduced remaining capacity rather than increased leakage of energy. To represent this, the idle drain amount is modeled as a constant power leakage rate that contributes to $\frac{dQ_u}{dt}$. Then, when the total charge used, Q_u , is normalized by the remaining battery capacity, Q_{ra} , the percent loss due to power leakage will be scaled by age.

It can be further shown that:

$$\begin{aligned} \frac{dQ_u}{dt} &= P_c + P_{idle} \\ \int_0^t \frac{dQ_u}{dt} dt &= \int_0^t (P_c + P_{idle}) dt \\ Q_u(t) &= (P_c + P_{idle}) \end{aligned} \quad (4)$$

This relationship is consistent with the standard definition of power.

$$P_C = (B \cdot S) \cdot z(t) + L + N \quad (5)$$

Here, $z(t)$ is an indicator that represents when the individual is actively using their phone.

$$z(t) = \begin{cases} 1, & \text{when screen is on} \\ 0, & \text{when screen is off} \end{cases} \quad (6)$$

$$L = L_{active} \cdot z(t) + L_0 \quad (7)$$

Processor load, L , is defined as a base value that is constantly applied and an active amount that increases the total when the phone is in use.

Taking the partial derivative $\frac{\partial SOC}{\partial t}$, it is shown that the rate of change of SOC with respect to time is the rate of battery usage scaled by $Q_{ra}(T)$.

$$\frac{\partial SOC}{\partial t} = -\frac{dQ_u}{dt} \cdot \frac{1}{Q_{ra}(T)} \quad (8)$$

This formula will be used to compute the time-to-empty predictions.

Table 3: Eq. 3-7 Variables/Parameters

Symbol	Description	Units
$z(t)$	Screen usage indicator	N/a
P_c	Power Consumption	W
P_{idle}	Idle Battery Drain	W
B	Screen Brightness	cd/m ²
S	Screen Area	m ²
L_{active}	Active Processor Load	W
L_0	Base Processor Load	W
N	Network Activity	W

The power consumption P_c is the rate at which the smartphone consumes energy from the battery and can be represented as a sum of contributing factors. These factors can be individually fluctuated to represent various usage patterns.

2.3 Degradation Due To Age and Temperature

To capture the impact of age and temperature on battery degradation, a recursive incremental model is proposed. At each cycle, it is assumed that all past battery history matters only through the current remaining battery capacity.

First, the relationship between temperature and capacity is approximated by statistical linear regression. Using data [4], two linear equations are used to characterize this behavior; one for temperatures above 22°C and one for those below. These equations are used in f_{temp} .

Since low temperatures are known to decrease capacity and higher temperatures are known to increase it, this is a suitable representation.

The equation f_{temp} calculates the amount of degradation of battery capacity per cycle at a specific temperature. The age of the battery is accounted for implicitly through the cycle counter, a . For example, when a smartphone's first cycle is at T_1 temperature, the value obtained using degradation factor f_{temp} is $Q_{r1} = Q_{r0} \cdot f_{temp}(T_1)$ where Q_{r0} is the base battery capacity. If the second cycle is at T_2 temperature, then the remaining battery capacity becomes $Q_{r2} = Q_{r1} \cdot f_{temp}(T_2)$. Therefore, at the a^{th} cycle at T_a temperature, the remaining battery capacity is $Q_{ra} = Q_{ra-1} \cdot f_{temp}(T_a)$.

2.4 Approximating Parameters

There is no public data on battery loss specifically for iPhones, so the value of P_{idle} is approximated using experiences from iPhone 16 users. Users report 2-4 % battery loss during idle time overnight from a full charge. [5]. We choose to approximate 'overnight' as 8 hours, and the phone's brand new capacity is 14.8Wh, thus we approximate $P_{idle} = \frac{14.8Wh \cdot 0.03}{8h} = 0.0555W$.

Brightness is measured in nits. One nit is one candela per square meter ($\frac{cd}{m^2}$). The typical maximum brightness of an iPhone 16 is 1000 nits, while the minimum is 1 nit [6]. The dimensions of an iPhone 16 are 2.82 in $\approx$ 0.0716 m width and 5.81 in $\approx$ 0.1476 m height [6]. Thus,

the area of an iPhone 16 is 0.01057 m^2 . To convert brightness and screen size into watts, lumens (lm) are needed. The formula for lumens is as follows [7]:

$$\text{Lumens} = \text{nits} \cdot \text{area} \cdot \pi \quad (9)$$

A typical smartphone can be treated as a Lambertian emitter [8], which allows the use of π in the formula. Thus, an iPhone 16 displays 33.21 lm at maximum brightness and 0.3321 lm at minimum brightness. A typical smartphone has a luminous efficacy ($\frac{\text{lm}}{\text{W}}$) ranging from 40 to 60 [9], and so a value of 50 will be used to represent the luminous efficacy of an iPhone 16. To convert lumens to watts:

$$\text{Watts} = \frac{\text{Lumens}}{\text{Luminous Efficacy}} \quad (10)$$

Therefore, the power an iPhone 16's screen gives off, in conjunction with brightness, ranges from 0.006642 W to 0.6642 W.

The iPhone 16's processing unit is the Apple A18 Pro, which is able to "briefly consume up to about 10 W when under high load, with average sustained power consumption figures hovering around 4 W" [10]. For daily use, a smartphone typically uses 0.5 – 2 W [11].

The power an iPhone uses from network activity ranges between 0.05 watts for a strong signal to 0.6 watts for a weak signal, while Wi-Fi continuously uses 0.03 watts [12].

3 Results

3.1 Realistic Usage Patterns

To understand how battery depletion is estimated, we first look at the behavior of individual parameters. To represent a realistic smartphone usage

pattern, $z(t)$ is created to indicate when the user is on their phone. Since the state of charge is indirectly affected by this value, it's important to understand what characterizes $z(t)$.

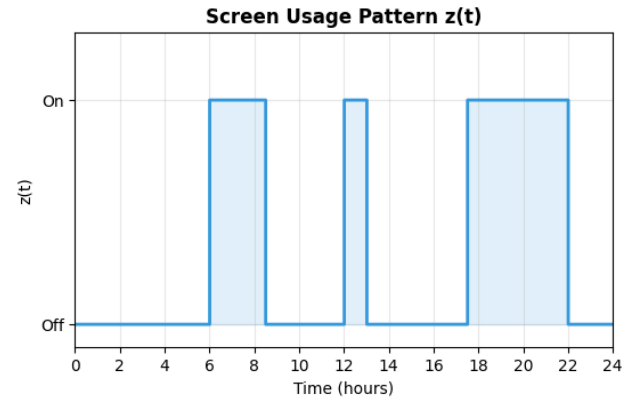


Figure 1: Phone Usage Indicator

Figure 1 represents the times that an individual is actively using their phone. In this instance, the day begins at 6 a.m. This block consists of any given morning routine, such as hygiene care and dressing up, and extends to 8 : 30a.m, where they are allowed to use their phone for the duration of their commute. From then until 12 p.m., they begin to work at their job, leading to $z(t)$ returning 0. Afterwards, there is an hour for lunch and then more work until their shift ends at 5 : 30 p.m. The next block consists of another commute and leisure in the evening until they decide to go to bed at 10 p.m. and start their day again at 6 in the morning.

This is a generic schedule of a smartphone that follows the typical work hours of a 9 to 5 job. Commute is described as the time it takes for the user to go from their current location to the next. Leisure implies that the user is able to freely be on their phone for an extended period of time. This is where one might see the greatest frequency of spikes in power and related variables. With these

terms explained, we can move forward with power consumption.

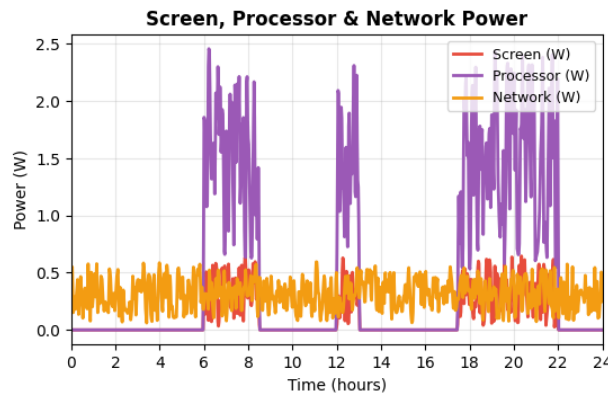


Figure 2: Power Consumption Breakdown

The power being used in the background is shown in figure 2. Here, the simplest plot to understand would be the wattage obtained from network activity; however, there are a few caveats that come with using cellular data. When the cellular connection is not transferring data, in other words, when the phone is idle, power from network activity remains low. While there is no network connection and the phone is still on cellular, it continuously uses the highest power trying to find a connection [12]. Multiple instances can be attributed to this behavior, with the simplest being represented by the plot, constantly generating values between the minimum and maximum power. The other two plots, "Screen" and "Processor", only return values when the phone is in use; hence, whenever $z(t) = 0$, B and L also return 0. Power usage from brightness remains fairly low even when in use, but it is non-negligible when considered over time. The processor load contributes the most wattage to power consumption. This parameter represents usage of streaming services, utility apps, games, etc. Each app has its own amount

of 'power consumption' that contributes to the total power used. Streaming services, for example, are high-load apps and can return values averaging 4 W [10], but more mundane apps, such as utilities, don't require a lot of power, allowing power consumption to remain at a consistent minimum [11].

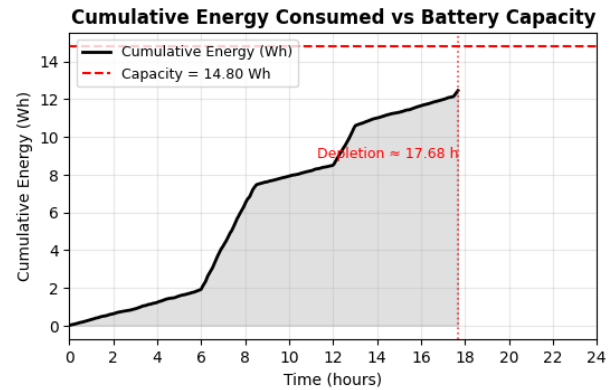


Figure 3: Cumulative energy over one cycle

Figure 3 plots the total energy used over time. This is the accumulation of all factors that contribute to power consumption, as well as taking into account the energy used from idle drain. The plot follows a similar pattern when compared to screen usage, with a few exceptions. During idle activity, power usage remains at a minimum, which translates to the rate of energy being used increasing at a seemingly linear rate. It appears linear because all values that go into power consumption are restricted to their respective ranges and accumulate over time. The actual shape of the plot consists of multiple hills and dips across the whole plane, reflecting the random behavior of power usage and the individual using the phone. During peak activity, the rate of energy consumption spikes. This pattern alternates until the battery is fully depleted.

Now, we consider the relationship between temperature and battery capacity.

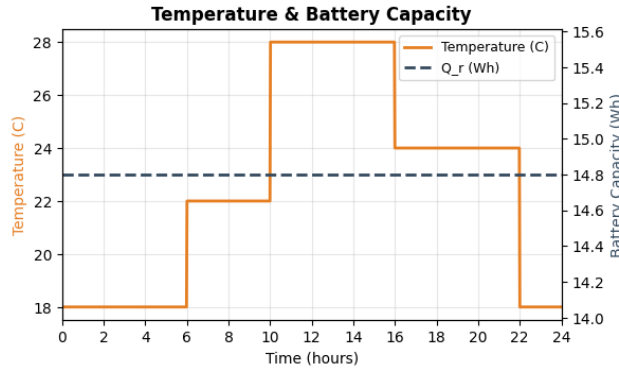


Figure 4: Temperature in relation to battery capacity over one cycle

The dual-axis chart plots temperature and battery capacity versus time. The construction of Q_{ra} recursively relies on itself and the function used for temperature. This function was obtained by regressing temperature to predict battery capacity using data sourced from Kaggle [4]. The regression model is f_{temp} , returning predicted battery capacity values based on the temperature of the current cycle and battery capacity of the previous cycle. While it sounds simple in nature, the behavior is slightly more complicated and is reflected by the plot. Battery capacity is reduced at lower temperatures, while at higher temperatures, it increases [13]. That's why even though the battery we chose to model has a capacity of 14.8 Wh, it peaks at almost 15.6 W when the temperature rises to 28° C which is equivalent to 82.4° F. This is a pseudo-maximum and, in turn, a pseudo-minimum when the temperature drops below the average. Thus, there is a direct relationship between temperature and battery capacity; however, this does not apply to the battery's life expectancy. In colder temperatures, chemical reactions are slower, making the battery less efficient at delivering energy. This can lead to a shorter lifespan for devices de-

pendent on that energy [13]. While that may be so, it is only temporary. High heat can cause permanent damage to the battery, effectively accelerating the aging process despite enhancing battery performance [13].

That being said, two separate models were created to better reflect this behavior. During any given cycle, the model used depends on the current temperature and is applied to the aging function Q_{ra} .

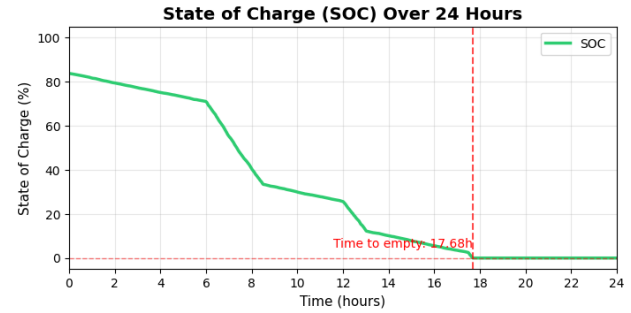


Figure 5: State of Charge over one cycle

Here, we see the state of charge over one cycle. In this cycle, the initial charge began at about 83%. Taking into account power consumption, along with idle drain, SOC decreases over time with various bursts of battery discharge when the phone is being actively used and may be under a high load. The time until the battery completely empties is determined to be 17.68 h. This can also be seen in figure 3. State of charge is just another representation of remaining energy. In figure 3, the battery depletes at about 12.3 Wh because it was not fully charged. We can also see that the plot follows the same general shape of cumulative energy, but inverted. Across one cycle, the aging process cannot be clearly seen, but becomes apparent over multiple cycles.

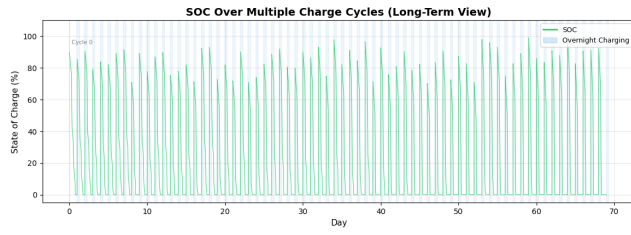


Figure 6: State of Charge over multiple cycles

Each "spike" in the graph represents a charge-discharge cycle with a varying initial charge. Over time, each cycle shortens due to the aging factor built into the model. The general pattern is consistent when viewing the system as a whole. This means that although one cycle's time-to-empty may take longer than the preceding cycle, the overall time it takes for the battery to completely drain is consistently decreasing.

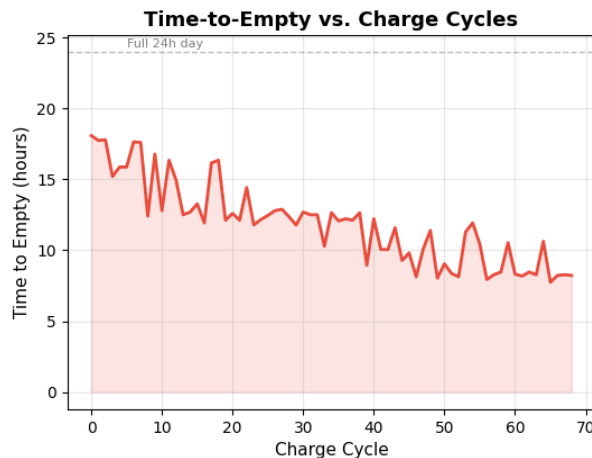


Figure 7: Time until empty over multiple cycles

The aforementioned pattern can be clearly seen in figure 7. Some cycles take longer than others to run empty, but ultimately, they reduce time-to-empty values.

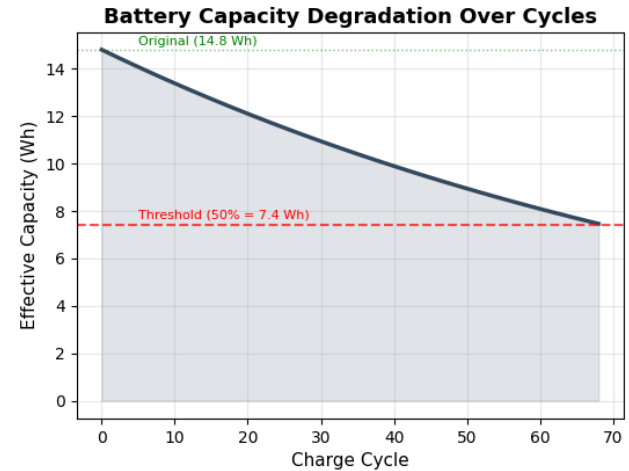


Figure 8: Continuous battery capacity degradation until termination

Figure 8 shows the degradation of the battery capacity over time. The plot showcases a gradual decline until it hits the set threshold of 50% of the initial battery capacity which results at the 69th cycle.

4 Discussion

4.1 Model Strengths and Limitations

The main strength of this model lies in its simplicity and logical structure. Based on observable relationships, the model produces conclusions that are both plausible and easy to interpret. All major contributing factors are explicitly accounted for, including battery usage patterns, temperature effects, and degradation history. Battery aging is incorporated through the recursive function Q_{ra} , which captures the cumulative impact of prior charge-discharge cycles. Furthermore, the model is not smartphone-specific and can be readily adapted to other battery-powered devices by re-evaluating the relevant parameter values.

Many assumptions and simplifications in the model creation limit the scope of our work and weaken the reliability of predictions using the model.

One key simplification is that the chemical components and the calendar age of the battery are not included in the degradation process. The general expectation is that older unused batteries are less effective than new unused batteries, but the model does not account for this behavior. While modeling the remaining battery capacity as a recursive incremental function is suitable for the stated assumptions, it restricts quantitative analysis approaches that require continuity.

Another limitation of the model is the assumption that ambient temperature is constant during each cycle. In reality, the ambient temperature fluctuates. If this model includes the impact of temperature change on *SOC*, conclusions may be drawn that support a user keeping the device colder or warmer to slow down the rate of *SOC* decline. Furthermore, it is expected that heavy processor load will raise the internal temperature of the phone, which may be shown to contribute to quicker *SOC* decline. However, this model treats temperature as an independent variable, assumed to be the ambient temperature of the current environment.

Lastly, Wi-Fi drains significantly less battery than cellular data [14]. Since the model only applies to devices using cellular data, the scope of application is limited.

4.2 Sensitivity Analysis

Local sensitivity analysis using the One-At-a-Time method is used to determine how realistic fluctuations in each variable impact *SOC*. These com-

putations serve to rank the parameters at the core of the model in order to determine which ones are worth changing. These parameters are screen brightness, processor load, network activity, and temperature. Idle battery drain rate and screen area are excluded as they are device-specific and not changeable by the user. The sensitivity index of a parameter will quantify the relative impact that fluctuations in the parameter value have on the *SOC*. To find the sensitivity indices, partial derivatives are computed and normalized as shown below.

Starting with the full model:

$$SOC(t, T) = SOC_0 - \frac{Q_u(t)}{Q_{ra}(T)} \quad (11)$$

$$Q_u(t) = (P_c + P_{idle}) \cdot t \quad (12)$$

$$Q_{ra}(T) = Q_{ra-1} \cdot f_{temp}(T) \quad (13)$$

$$P_c = (B \cdot S) \cdot z(t) + L + N \quad (14)$$

$$z(t) = \begin{cases} 1, & \text{screen on} \\ 0, & \text{screen off} \end{cases} \quad (15)$$

$$L = L_{active} \cdot z(t) + L_0 \quad (16)$$

Now, substitute (12), (13), (14), and (16) into (11): Note: fractions are separated to improve readability and multiplication signs are removed due to space constraints.

$$SOC(t, T) = SOC_0 - \frac{(BS)z(t)}{Q_{ra-1}f_{temp}(T)} t - \frac{L_{active}z(t) + L_0}{Q_{ra-1}f_{temp}(T)} t - \frac{N + P_{idle}}{Q_{ra-1}f_{temp}(T)} t \quad (17)$$

Now, compute the first partial derivatives of *SOC* with respect to *B*, *L*, *N*, and *T*

$$\frac{\partial SOC}{\partial B} = -\frac{Sz(t)t}{Q_{r_{a-1}}f_{temp}(T)} \quad (18)$$

$$\frac{\partial SOC}{\partial L} = -\frac{t}{Q_{r_{a-1}}f_{temp}(T)} \quad (19)$$

$$\frac{\partial SOC}{\partial N} = -\frac{t}{Q_{r_{a-1}}f_{temp}(T)} \quad (20)$$

$$\frac{\partial SOC}{\partial T} = \left(\frac{P_c + P_{idle}}{Q_{r_{a-1}}} \right) t \frac{f'_{temp}(T)}{f_{temp}(T)^2} \quad (21)$$

Note: $\frac{\partial SOC}{\partial T}$ is collapsed due to space constraints and lack of readability. Additionally, L is treated as an independent parameter for simplicity. Now, normalizing each partial gives its sensitivity index, denoted by α .

$$\alpha_B = \frac{B}{SOC} \cdot -\frac{Sz(t)t}{Q_{r_{a-1}}f_{temp}(T)} \quad (22)$$

$$\alpha_L = \frac{L}{SOC} \cdot -\frac{t}{Q_{r_{a-1}}f_{temp}(T)} \quad (23)$$

$$\alpha_N = \frac{N}{SOC} \cdot -\frac{t}{Q_{r_{a-1}}f_{temp}(T)} \quad (24)$$

$$\alpha_T = \frac{T}{SOC} \cdot \frac{P_c + P_{idle}}{Q_{r_{a-1}}} t \cdot \frac{f'_{temp}(T)}{f_{temp}(T)^2} \quad (25)$$

Suppose parameters correspond to typical smartphone conditions. For this analysis, we assume the battery has a remaining battery capacity of 10Wh and is being evaluated at 1 hour. The screen brightness, processor load, network activity, and screen area correspond to the iPhone 16 under realistic conditions. The degradation factor and its derivative are estimated.

Table 4: Baseline Parameter Values

Parameter	Symbol	Value
State of Charge	SOC	95%
Screen Brightness	B	500 nits
Active Processor Load	L_{active}	2 W
Base Processor Load	L_0	0.5 W
Network Activity	N	0.05 W
Temperature	T	21 °C
Screen Area	S	0.01057 m ²
Screen "On" Indicator	$z(t)$	1
Idle power Drain	P_{idle}	0.0555W
Evaluation time	t	1 h
Rem. Battery Capacity	$Q_{r_{a-1}}$	10 Wh
Degradation Factor	$f_{temp}(T)$	0.99
Deriv. of Degradation Factor	$f'_{temp}(T)$	-0.002

Using these values, the partial derivatives are approximated as:

$$\frac{\partial SOC}{\partial B} \approx -0.001067$$

$$\frac{\partial SOC}{\partial L} \approx -0.101010$$

$$\frac{\partial SOC}{\partial N} \approx -0.101010$$

$$\frac{\partial SOC}{\partial T} \approx -0.001610$$

Note: Units are not specified because only the magnitude and sign are needed.

Therefore, the normalized sensitivity indices are approximately:

$$\alpha_B \approx \frac{500}{0.95} \cdot (-0.001067) \approx -0.5616$$

$$\alpha_L \approx \frac{2.5}{0.95} \cdot (-0.101010) \approx -0.2658$$

$$\alpha_N \approx \frac{0.05}{0.95} \cdot (-0.101010) \approx -0.0053$$

$$\alpha_T \approx \frac{21}{0.95} \cdot (-0.001610) \approx -0.0356$$

The magnitude of a sensitivity index quantifies the relative impact of parameters on changes in *SOC* for the chosen baseline values, and the sign indicates whether the parameter is directly or inversely related to *SOC*. All of the indices are negative, which is consistent with the expectation that increasing brightness, processor load, network activity, or temperature will decrease *SOC*.

When the screen is on, brightness is the most sensitive parameter, meaning increasing brightness decreases *SOC* more quickly than the other parameters. Processor load is the next most sensitive, decreasing *SOC* faster than temperature and network load, but still slower than brightness. Temperature has a small indirect effect, slightly decreasing *SOC* due to increasing temperature decreasing the remaining capacity. Network activity has the smallest impact on *SOC*.

To extend the analysis, normalized sensitivity indices are computed when the screen is off, $z(t) = 0$. Thus the partial derivatives are:

$$\begin{aligned}\frac{\partial SOC}{\partial B} &= 0 \\ \frac{\partial SOC}{\partial L} &\approx -0.101010 \\ \frac{\partial SOC}{\partial N} &\approx -0.101010 \\ \frac{\partial SOC}{\partial T} &\approx -0.000124\end{aligned}$$

And the normalized sensitivity indices are:

$$\begin{aligned}\alpha_B &= \frac{500}{0.95} \cdot (0) = 0 \\ \alpha_L &\approx \frac{0.5}{0.95} \cdot (-0.101010) \approx -0.0531 \\ \alpha_N &\approx \frac{0.05}{0.95} \cdot (-0.101010) \approx -0.0053 \\ \alpha_T &\approx \frac{21}{0.95} \cdot (-0.000124) \approx -0.0027\end{aligned}$$

Again, all of the nonzero indices are negative,

which is consistent with the expectation that increasing brightness, processor load, network activity, or temperature will decrease *SOC* regardless of device activity.

When the screen is off, brightness is no longer a contributor to change in *SOC*, so its sensitivity is zero. Since the formula of $\frac{\partial SOC}{\partial T}$ (21) depends on total power (which $z(t)$ affects by contributing screen usage and active processor load), the sensitivity index α_T is greatly decreased when the screen is off. The ranking has changed, with *SOC* becoming most sensitive to changes in processor load, less sensitive to network activity, and least sensitive to the indirect impact of temperature.

Although this sensitivity analysis considers only two scenarios, others can be computed using the same method, and interpretations of those sensitivity indices could be a point of extension in future work. It is also important to note that these rankings are local to the chosen baseline values and may change under different conditions.

5 Recommendations

5.1 Limiting Behavior

There are quite a few changes a cellphone user can make to yield the largest improvement to battery life, such as using low brightness whenever possible. Keeping brightness at a minimum yields the greatest effect on *SOC* since the lowest values offer very minimal power usage. When it comes to minimizing processor load, there aren't too many realistic solutions besides limiting the use of high-load applications. While using a smartphone for something such as streaming services or video-calling leads to faster battery drain, avoiding these

activities diminishes the value of owning a smart-phone for many users. So, although this is a viable solution, it may not be practical as a general recommendation to prolong battery life. Power usage from network activity is the same way, meaning there are ways to reduce its consumption, but they are not always possible. It would be best to always stay connected to Wi-Fi as well as turn off cellular data, but not everywhere provides Wi-Fi service, and even fewer are freely available. Not only that, but there are multiple internet service providers whose area of effectiveness is limited. Thus, staying within that area would be best as it would keep the power from data transfers at a minimum. Influence from temperature is the most severe since constant exposure to high temperatures can cause permanent damage to a phone battery. The most practical advice would be to limit phone use outside and in unconditioned spaces.

5.2 Low Power Mode

Although the recommended changes detailed in the previous subsection are great, the average user might not understand how to apply them. To bridge this gap, most battery-powered devices have a "Low Power Mode" setting. Using this mode, devices usually limit things like brightness and cellular usage (by prioritizing conservation over quickest speed) as well as completely disabling processes like background app refresh and mail fetch. [15].

6 Future Work

A common piece of advice from smartphone manufacturers is to avoid charging to full capacity and

draining the device till it dies. Future work could include this behavior in the model to better reflect real-world usage.

In this model, battery degradation is simplified and relies just on cycle count and temperature per cycle. In reality, a battery can degrade due to physical, electrical, or chemical factors [16]. A future model might replace the incremental recursive function Q_{ra} with a continuous-time version that accounts for these additional mechanisms.

The parameters of this model could also be vastly improved with more model's robustness, enabling it to account for various unique background tasks like location services and email fetch. Global sensitivity analysis could also uncover other interactions between parameters under various other usage scenarios.

7 Conclusion

In modern electronic devices, lithium-ion batteries play a critical role in enabling continuous operation. Most people consider smartphones essential for everyday life. understanding the ins-and-outs of lithium-ion batteries is essential. This work develops a simplified yet realistic model for estimating the state of charge based on key influencing factors, including screen brightness, processor load, network activity, temperature, and battery usage history. Through sensitivity analysis, the relative impact of these factors is used to rank them according to which caused the greatest effect on *SOC*.

This work is significant because it provides a simple framework that relates many factors to *SOC*. The model provides insights into the role of temperature in battery degradation. Lastly, the

model can be used to predict the time-to-empty. To demonstrate prediction power, our model uses randomly generated values for each parameter within their realistic ranges, modeling how a user's smart-phone usage varies day-to-day. Additionally, select parameters can be adjusted to apply the model to different devices.

8 References

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9 AI Use Report

1. Anthropic *Claude* (Jan 31, 2026 version, Sonnet 4.5)
Auto-completions for code used in preparing our models.
2. OpenAI *ChatGPT* (Jan 31, 2026 version, ChatGPT-5.2)
Fixing LaTeX syntax.
3. OpenAI *ChatGPT* (Feb 1, 2026 version, ChatGPT-5.2)
Uploaded passages to remove and replace informal language.
4. Superhuman Platform Inc. *Grammarly* (Feb 1, 2026 version)
Grammar and punctuation assistance.